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and collagen fibres which gives strength to the bones. It is the main tissue that provides structural frame to the body. Bones support and protect softer tissues and organs. The bone cells (osteocytes) are present in the spaces called lacunae. Limb bones, such as the long bones of the legs, serve weight-bearing functions. They also interact with skeletal muscles attached to them to bring about movements. The bone marrow in some bones is the site of production of blood cells.

Blood is the fluid connective tissue containing plasma, red blood cells (RBC), white blood cells (WBC) and platelets. It functions as the transport medium for the cardiovascular system, carrying nutrients, wastes, respiratory gases throughout the body. You will learn more about blood in Chapter 7.

Important connective tissue disorders: (Heritable types)

1. **Ehler's -Danlos syndrome** – Defect in the synthesis of collagen in the joints, heart valves, organ walls and arterial walls.
2. **Stickler syndrome** – Affects collagen and results in facial abnormalities.
3. **Rhabdomyosarcoma** – Life threatening soft tissue tumour of head, neck and urinogenital tract.

Autoimmune connective tissue disorders

1. **Rheumatoid arthritis:** The immune cells attack and inflame the membranes around the joints. It can also affect heart, lungs and eyes.
2. **Sjogren's syndrome:** Progressive inability to secrete saliva and tears.

Biopsy is an examination of tissue or liquid removed from a living body to discover the presence, cause or extent of a disease.

Autopsy is a post-mortem (dissection of a dead body) examination to discover the cause of death or the extent of disease.

The field of **Forensic science** effectively uses the histological techniques to trace out crimes.

3.4 Muscle Tissue

Each muscle is made of many long, cylindrical fibres arranged in parallel arrays. These fibres are composed of numerous fine fibrils, called myofibrils. Muscle fibres contract (shorten) in response to stimulation, then relax (lengthen) and return to their uncontracted state in a coordinated fashion. In general muscles play an active role in all the movements of the body.

Muscles are of three types, skeletal, smooth and cardiac. **Skeletal muscle tissue** is closely attached to skeletal bones. In a typical muscle such as the biceps, the striated (striped) skeletal muscle fibres are bundled together in a parallel fashion. A sheath of tough connective tissue encloses several bundles of muscle fibres (You will learn more about this in Chapter 9).

The **smooth muscle** fibres taper at both ends (fusiform) and do not show striations (Figure 3.7). Cell junctions hold them together and they are bundled together in a connective tissue sheath. The walls of internal organs such as the blood vessels, stomach and intestine contain this type of muscle